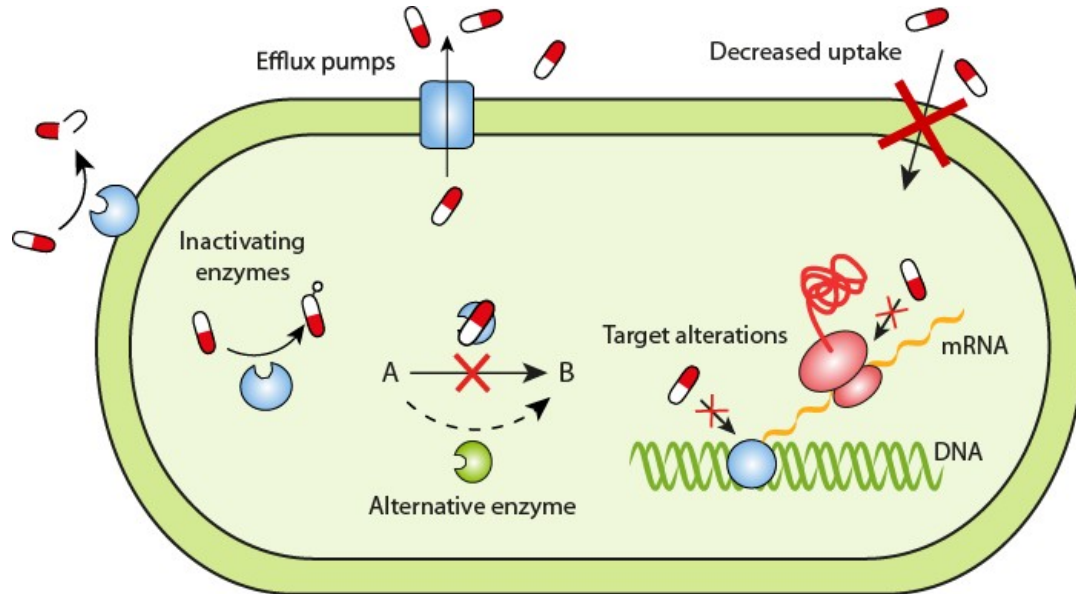




AMR Pipeline and Sample Report

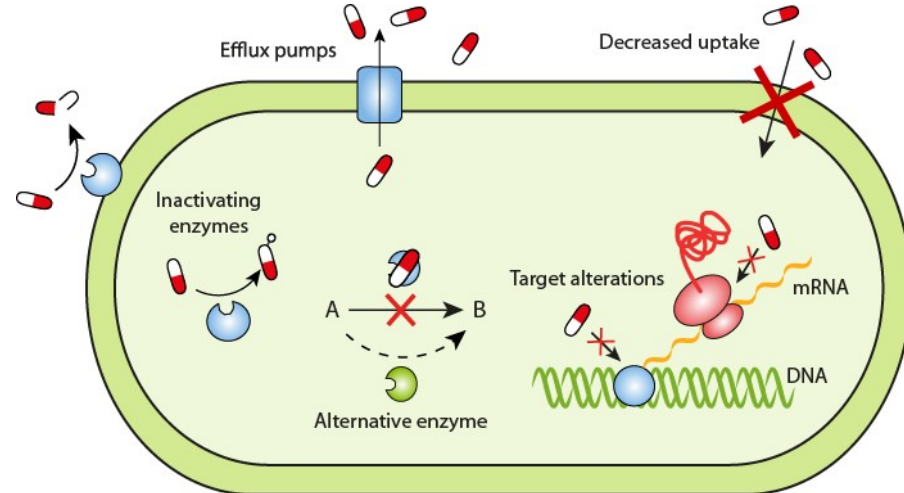
Antimicrobial Resistance (AMR) Mechanisms

- Stop antibiotic from reaching target
- Modify or bypass the antibiotic target



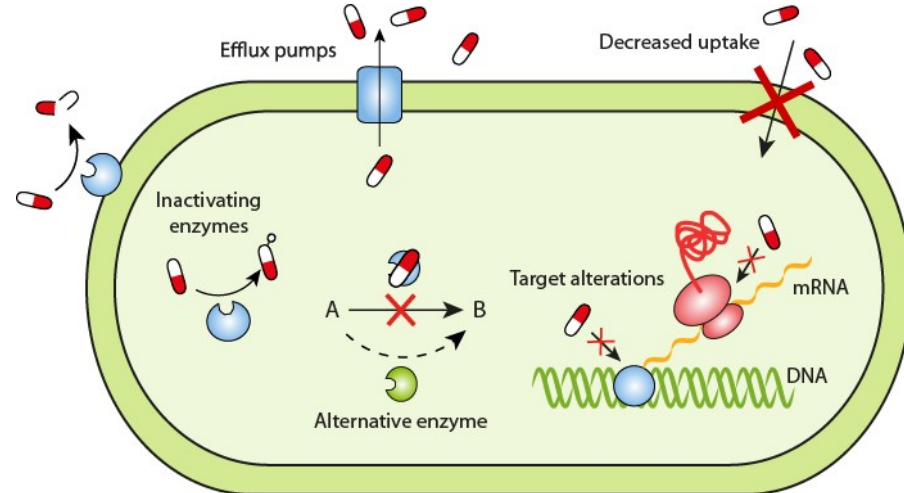
Antimicrobial Resistance (AMR) Mechanisms

- Stop antibiotic from reaching target
 - Pump the antibiotic out of the cell (efflux pumps)
 - Decrease uptake (membrane permeability)
 - Destroy or modify the antibiotic (inactivating enzymes)



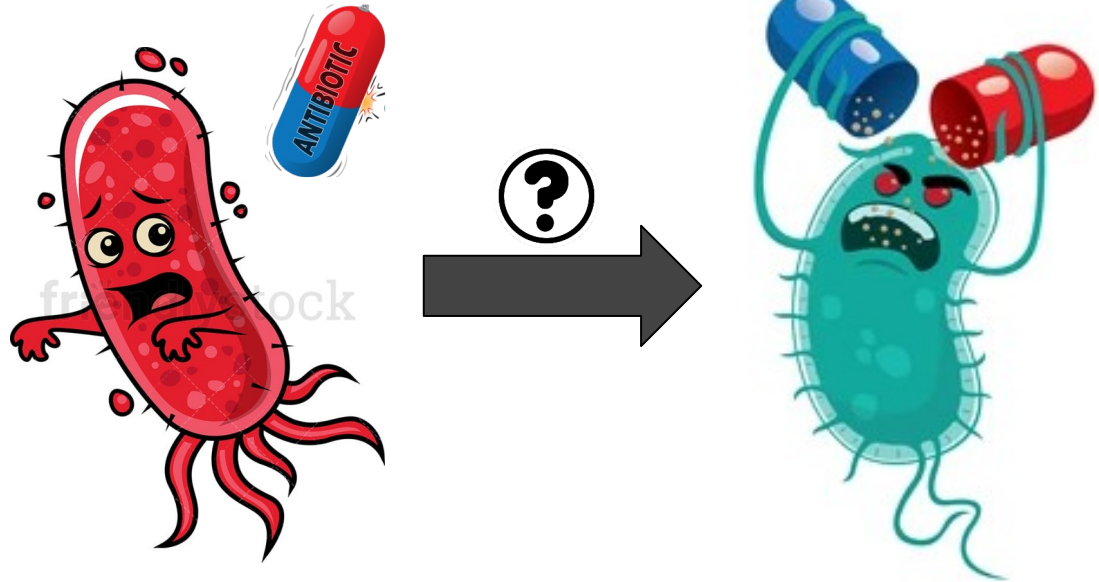
Antimicrobial Resistance (AMR) Mechanisms

- Stop antibiotic from reaching target
 - Pump the antibiotic out of the cell (efflux pumps)
 - Decrease uptake (membrane permeability)
 - Destroy or modify the antibiotic (inactivating enzymes)
- Modify or bypass the antibiotic target
 - Camouflage or reprogram the target (target alteration)
 - Use a different protein (expression of alternative proteins)

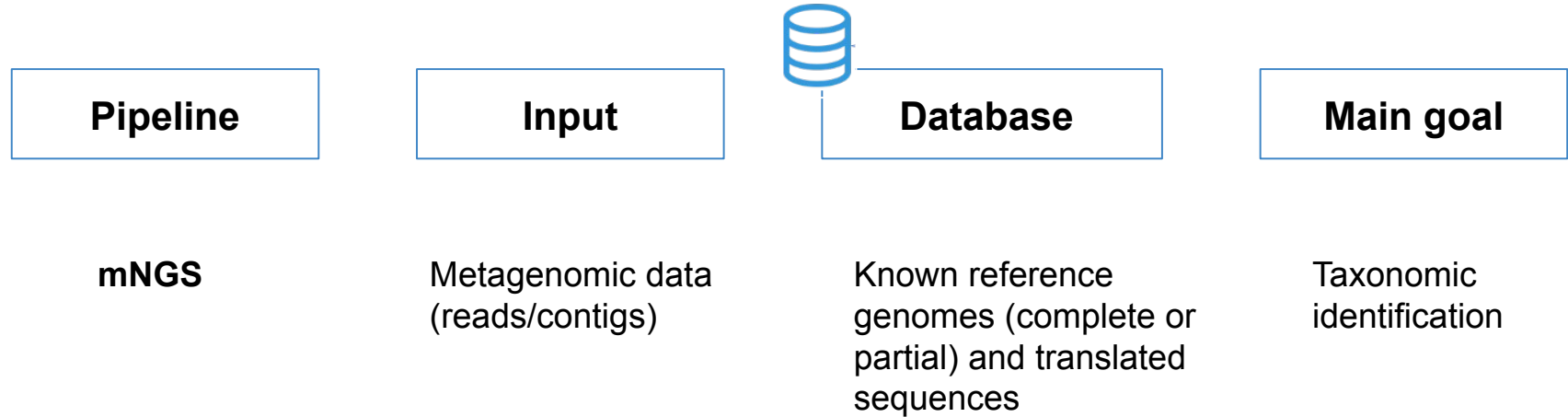


What can give rise to AMR?

- Intrinsic resistance (e.g., Gram positive vs Gram negative)
- Acquisition of new genetic material (Horizontal gene transfer)
- Mutation



mNGS vs AMR Detection Pipelines



mNGS vs AMR Detection Pipelines

Pipeline	Input	 Database	Main goal
mNGS	Metagenomic data (reads/contigs)	Known reference genomes (complete or partial) and translated sequences	Taxonomic identification
AMR	Metagenomic or whole genome sequence data (reads/contigs)	Known reference AMR genes and translated sequences	AMR gene detection

CZ ID's AMR Pipeline Highlights

- Supports whole genome sequence (WGS) and metagenomic data.
- Uses the [Comprehensive Antibiotic Resistance Database \(CARD\)](#).
 - Combines the Antibiotic Resistance Ontology (ARO) with curated AMR gene sequences and resistance-conferring mutations.
- Uses both sequence reads and assembled sequences (contigs) for AMR gene detection.
- Performs pathogen-of-origin prediction by matching query sequences to known pathogen sequences found in CARD (beta).

AMR Database

CARD includes two database modules:

1. Canonical CARD: Includes curated and functionally characterized AMR genes from the literature

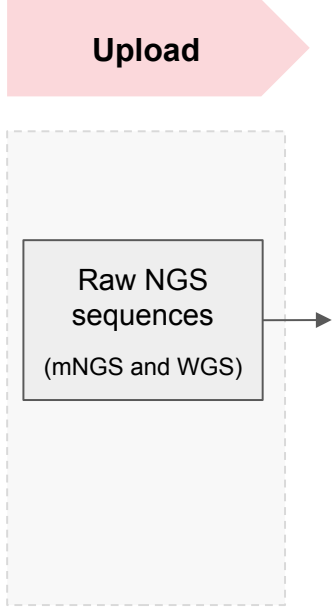
2. Resistomes and Variants (aka. **WildCARD**): Supplemental database of computationally predicted AMR genes and variants and their Prevalence.

- Expands CARD, which is strongly biased towards common clinical pathogens
- Only used for pathogen-of-origin prediction

AMR Pipeline Workflow

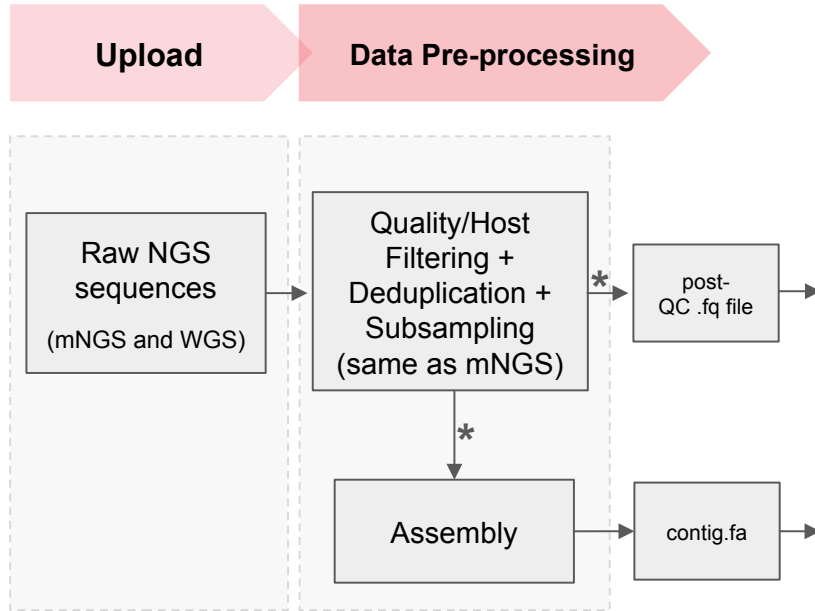
Upload

Raw NGS
sequences
(mNGS and WGS)



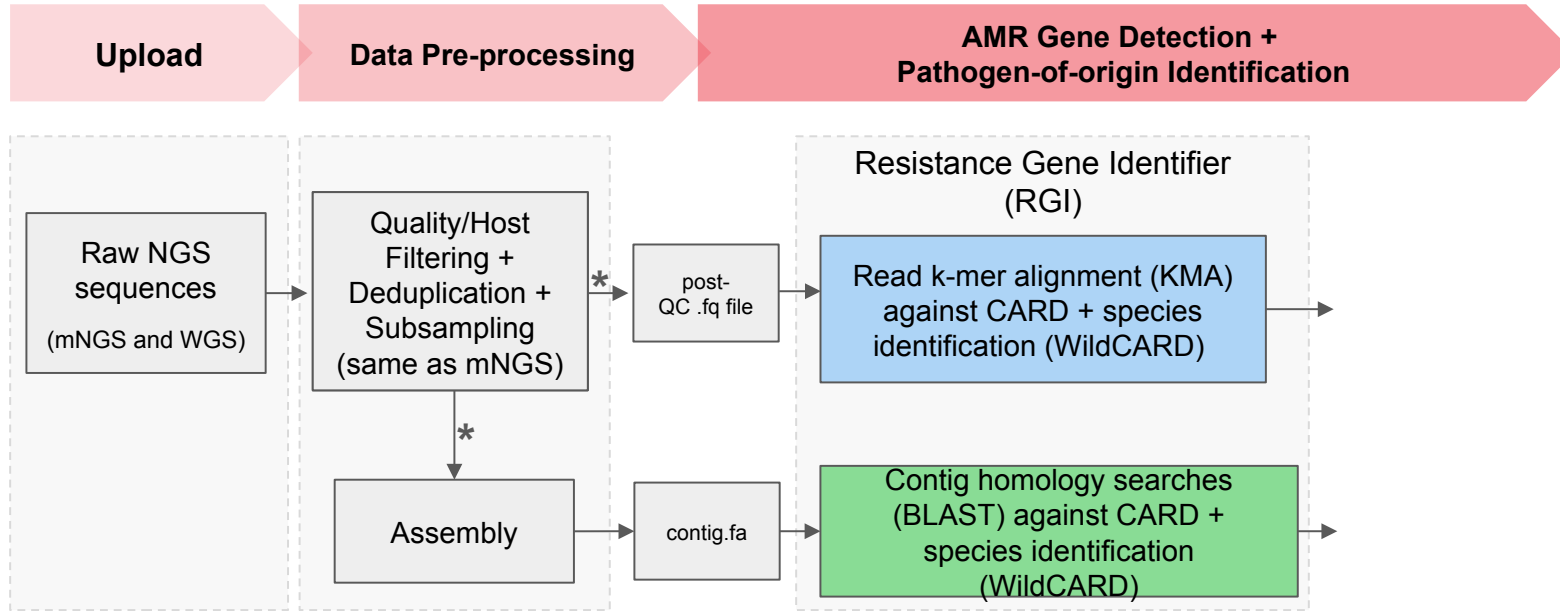
```
graph LR; Upload[Upload] --> RawNGS[Raw NGS sequences (mNGS and WGS)]; RawNGS --> Next[ ];
```

AMR Pipeline Workflow



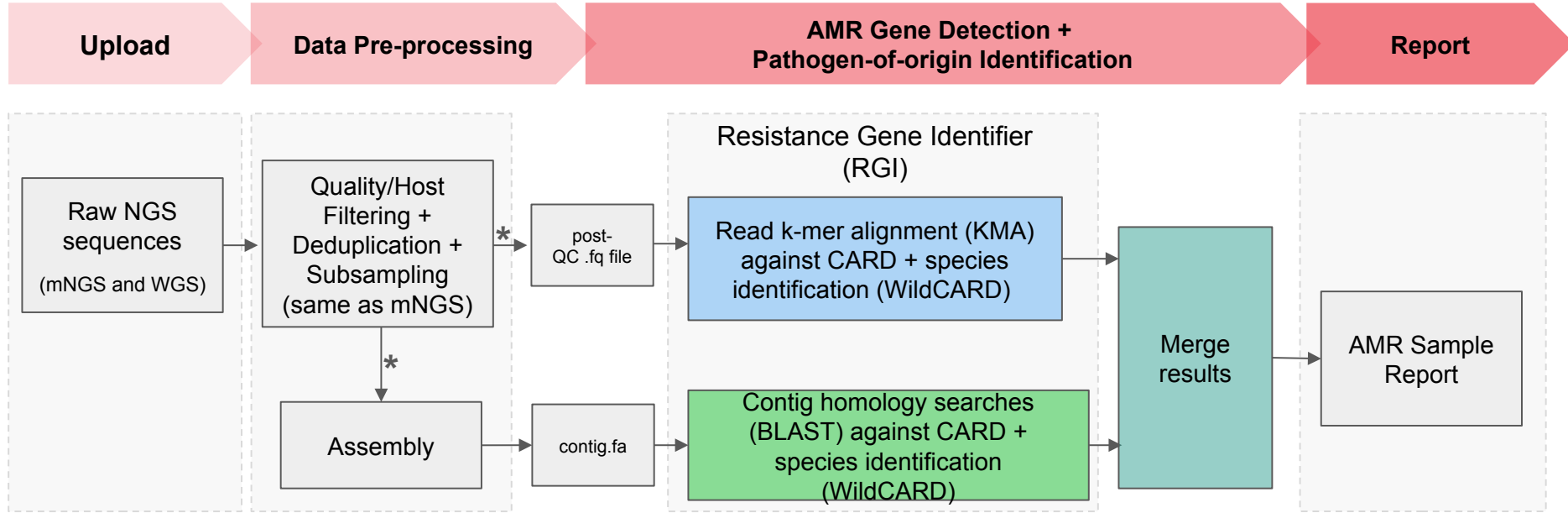
*Duplicate reads are added back before contig assembly and AMR gene detection

AMR Pipeline Workflow



*Duplicate reads are added back before contig assembly and AMR gene detection

AMR Pipeline Workflow



*Duplicate reads are added back before contig assembly and AMR gene detection

AMR Sample Report

< mNGS_Demo

AMR Pipeline v1.2.15 | CARD DB: 3.2.6 | processed 34 minutes ago | [SAMPLE DETAILS](#)

AMR_test1_reads_nh ▾

[Share](#) [Download ▾](#) [?](#) [...](#)

Metagenomic Antimicrobial Resistance



38 Rows

Gene ^	Drug Class	High-level Drug Class	Mechanism	Model	» Contigs				» Reads					
					Contigs	Cutoff	%Cov	%Id	Reads	rPM	%Cov	Cov. Depth	dPM	
AAC(6')-Ib7	aminoglycoside antibiotic	aminoglycoside antibiotic	antibiotic inactivation	protein homolog	0	-	-	-	4	1.81	24.05	0.56	0.25	
aadA10	aminoglycoside antibiotic	aminoglycoside antibiotic	antibiotic inactivation	protein homolog	0	-	-	-	2	0.9	24.22	0.27	0.12	
aadA27	aminoglycoside antibiotic	aminoglycoside antibiotic	antibiotic inactivation	protein homolog	0	-	-	-	1	0.45	13.31	0.14	0.06	

AMR Sample Report: Gene Information

< mNGS_Demo

AMR_test1_reads_nh

MetagenomicAntimicrobial Resistance

38 Rows

GeneDrug ClassHigh-level Drug ClassMechanismModel

»Contigs»Reads

ContigsCutoff%Cov%IdReadsrPM%CovCov. DepthdPM

- **Gene:** Gene name for best match in CARD
- **Gene Family:** Refers to the ARO category for gene family
- **Drug Class:** Refers to the ARO category specifying resistance to a given antibiotic molecule, including antibiotic/adjuvant combination medications.
- **High-level Drug Class:** Refers to the antibiotic family of identified drug class.
- **Resistance Mechanism:** Refers to the ARO category specifying the resistance mechanism for a given AMR gene.
- **Model:** Specifies the model used for AMR detection. Models distinguish underlying molecular categories for detected AMR sequences (e.g., presence of genes vs mutations). More on models on the next slide...

Models for AMR Gene Detection

- **Protein Homolog Models (PHM):** Detect dedicated AMR genes (“presence/absence”). Example: [NDM-1](#), a metallo-beta-lactamase isolated from *Klebsiella pneumoniae* with nearly complete resistance to all beta-lactam antibiotics
 - Represent $\sim\frac{2}{3}$ of CARD
 - Contigs: BLASTP against CARD database (protein:protein alignment).
 - Reads: KMA against PHM sequences (**only model used for calling reads**)

Models for AMR Gene Detection

- **Protein Homolog Models (PHM):** Detect dedicated AMR genes
- **Protein Variant Models (PVM):** Detect AMR acquired via mutation of house-keeping genes or antibiotic targets.
 - 2nd largest in CARD
 - PVMs screen query sequences for curated sets of mutations that could differentiate them from antibiotic susceptible genotypes. Example: [gyrA](#), mutations in the A subunit of DNA gyrase reduce its affinity for fluoroquinolones.

Query	7	SELNHCDECFALMNPLMILVKIIKLRWIIHILSYDQMVSRKINNQTTRCANSIFYTTLFTN	66
		\$+ N C+ FALMNPLMILVKIIKLRWIIHILSYDQMVSRKINNQTTRCANSIFYTTLFTN	
Sbjct	7	SQSNICNRDFALMNPLMILVKIIKLRWIIHILSYDQMVSRKINNQTTRCANSIFYTTLFTN	66
Query	67	SAPNYT	72
		SAPNYT	
Sbjct	67	SAPNYT	72

Models for AMR Gene Detection

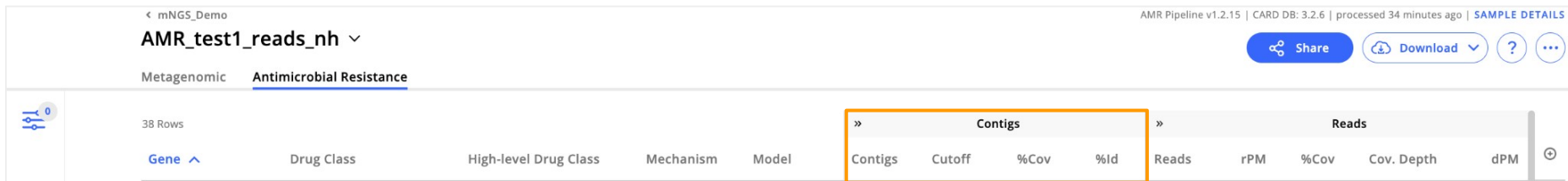
- **Protein Homolog Model (PHM)**- Detect dedicated AMR genes
- **Protein Variant Model (PVM)** - Designed to detect AMR acquired via mutation of house-keeping genes or antibiotic targets
- **Protein Overexpression Models (POM)** - Detect mutations in regulatory proteins.
 - Similar to PVM but restricted to regulatory proteins.
 - POM screen for mutations that may lead to overexpression of efflux complexes.
 - Example: [MexS](#), a suppressor of MexT, which is an activator of the multidrug pump MexEF-OprN. Mutations in MexS lead to multidrug resistance.

Models for AMR Gene Detection

- **Protein Homolog Model (PHM)**- Detect dedicated AMR genes
- **Protein Variant Model (PVM)** - Detect AMR acquired via mutation of house keeping genes or antibiotic targets.
- **Protein Overexpression Model (POM)** - Detect mutations in regulatory proteins.
- **rRNA Gene Variant Models (RVM)** - Detect AMR acquired via mutation of genes encoding rRNA (nucleotide:nucleotide alignment).
 - Screens query sequences for curated sets of mutations that could differentiate them from antibiotic susceptible genotypes
 - Example: [Cjej_23S_ERY](#), point mutations in the 23S ribosomal RNA domain of *C. jejuni* which confer resistance to macrolide antibiotics, including erythromycin.

GAGGTGACGATTACCTTTCAATCGG
|||||
GAGGTGACGATTACCTAACAATCGG

AMR Sample Report: Contig Metrics



AMR Pipeline v1.2.15 | CARD DB: 3.2.6 | processed 34 minutes ago | [SAMPLE DETAILS](#)

AMR_test1_reads_nh ▾

Metagenomic Antimicrobial Resistance

38 Rows

Contigs				Reads				
Contigs	Cutoff	%Cov	%Id	Reads	rPM	%Cov	Cov. Depth	dPM

- **Contigs:** Total number of contigs matching a given AMR gene
- **Cutoff:** Cutoff used to detect AMR-associated contigs based on curated BLAST bit-score thresholds.
 - **Perfect:** Detects perfect or identical matches to the curated reference sequences in CARD
 - **Strict:** Detects previously unknown variants of known AMR genes and includes a secondary screen for key mutations (predicts functional variants)
 - **Loose:** Detects new and more distant homologs of AMR genes by working outside of established BLAST bit-score cut-offs allowing for AMR gene discovery. CZ ID only reports Loose contig matches that have at least 95% identity to known AMR genes (identified as "Nudged")
 - Nudged cut off does NOT account for alignment length

(More info [here](#))

AMR Sample Report: Contig Metrics (cont.)

← mNGS_Demo

AMR Pipeline v1.2.15 | CARD DB: 3.2.6 | processed 34 minutes ago | [SAMPLE DETAILS](#)

AMR_test1_reads_nh ▾

[Share](#) [Download ▾](#) [?](#) [...](#)

Metagenomic Antimicrobial Resistance

38 Rows

[Gene ▴](#)

Drug Class

High-level Drug Class

Mechanism

Model

» Contigs

» Reads

Contigs	Cutoff	%Cov	%Id	Reads	rPM	%Cov	Cov. Depth	dPM
---------	--------	------	-----	-------	-----	------	------------	-----

- **Contig Coverage Breath (% Cov):** Percentage length of the reference sequence that was covered by contigs
- **Contig Percent Identity (% ID):** Average percent identity between contigs and their top match in CARD
- **Contig Species:** Pathogen-of-origin prediction based on AMR-associated contig sequences.

AMR Sample Report: Read Metrics

AMR Pipeline v1.2.15 | CARD DB: 3.2.6 | processed 34 minutes ago | [SAMPLE DETAILS](#)

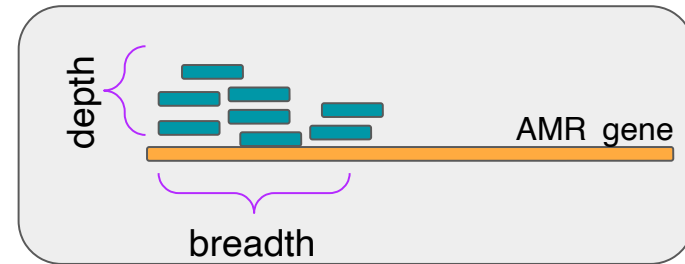
AMR_test1_reads_nh ▾

Metagenomic Antimicrobial Resistance

38 Rows

Contigs					Reads				
Contigs	Cutoff	%Cov	%Id		Reads	rPM	%Cov	Cov. Depth	dPM

- **Reads:** Total number of reads mapping to a given AMR reference sequence.
- **Read Coverage Breadth (% Cov):** Percent length of the reference sequence covered by read sequences.
- **Read Coverage Depth (Cov. Depth):** Mean read depth across the reference sequence.
- **Read Species:** Pathogen-of-origin prediction based on AMR-associated reads.



Normalized Read Metrics

When comparing across samples is important to **normalize** the data to make sure what we see is not just an artifact of one sample having more reads than another.

- **Reads per million (rPM)**

$$\text{rPM} = \frac{\text{Reads matching gene or allele}}{(\text{Total reads} - \text{ERCC reads}) \times \text{Subsampled fraction}} \times 10^6$$

- **Depth per million (dPM)**

$$\text{dPM} = \frac{\left(\frac{\text{Bases mapped to gene or allele}}{\text{Length of gene or allele}} \right)}{(\text{Total reads} - \text{ERCC reads}) \times \text{Subsampled fraction}} \times 10^6$$

Applying Filters to AMR Sample Report

AMR genes tend to be at low abundances unless they are enriched (e.g., FLASH protocol). Use relaxed parameters compared to filters used for mNGS analysis

- Suggested threshold:
 - AMR genes with at least 5% (mNGS data) or 50% (WGS data) coverage.
 - At least 5 reads matching AMR gene.

AMR Pipeline v1.2.14 | CARD DB: 3.2.6 | processed 20 hours ago | [SAMPLE DETAILS](#)

AMR_test1_reads_nh

Antimicrobial Resistance

38 Rows

» Contigs

» Reads

Gene	Drug Class	High-level Drug Class	Mechanism	Model	Contigs	Reads	rPM	%Cov	Cov. Depth	dPM
tb7	aminoglycoside antibiotic	aminoglycoside antibiotic	antibiotic inactivation	protein homolog	0	4	1.81	24.05	0.55	0.25
aadA10	aminoglycoside antibiotic	aminoglycoside antibiotic	antibiotic inactivation	protein homolog	0	2	0.9	24.22	0.27	0.12
aadA27	aminoglycoside antibiotic	aminoglycoside antibiotic	antibiotic inactivation	protein homolog	0	1	0.45	13.31	0.14	0.06

Filters

THRESHOLDS

DRUG CLASS

Available filters

Filters

THRESHOLDS

Configure Thresholds

Number of Contigs

Contigs % Coverage

Contigs % Identity

Number of Reads

rPM (Reads per Million)

Reads % Coverage

Reads Coverage Depth

dPM (Depth per Million)

Available metrics to filter table

Downloading Data from AMR Sample Report

AMR-v2-test-final

RR134e_00014_S14_357153_reads_nh

AMR Pipeline v1.2.3 | CARD DB: 3.2.3 | processed a day ago | [SAMPLE DETAILS](#)

ShareDownload

?

...

Antimicrobial Resistance

17 Rows

Gene	Drug Class	Mechanism	Model	Contigs				Reads			
				Contigs	Cutoff	%Cov	%Id	Reads	rPM	%Cov	Cov
arlR	disinfecting agents and antiseptics;...	antibiotic efflux	protein homolog	1	Perfect	99.55	100	494	239.58	100.0	89
arlS	disinfecting agents and antiseptics;...	antibiotic efflux	protein homolog	1	Strict	99.78	99.78	799	387.5	100.0	66.65 32.32
IMP-46	carbapenem; cephalosporin;...	antibiotic inactivation	protein homolog	0	-	-	-	41	19.88	9.37	0.14 0.07

Download Report Table (.csv)
Download Report Table with Applied Filters (.csv)
Download Non-Host Reads (.fasta)
Download Non-Host Contigs (.fasta)
Download Comprehensive AMR Metrics File (.tsv)
Download Intermediate Files (.zip)

Download dropdown menu

AMR Sample Report Download Options

File name	File type	Description
Report Table	CSV	Primary AMR report summarizing detected AMR gene information, quality of sequence matches, and pathogen-of-origin prediction results
Report Table with Applied Filters	CSV	Primary AMR report for remaining AMR genes after applying filters
Non-host Reads	FASTA	Quality-filtered reads used for AMR analysis after filtering out host and human sequences Note: Paired-end data will be merged into a single FASTA file.
Non-host Contigs	FASTA	Assembled contig sequences used for AMR analysis
Comprehensive AMR Metrics	TSV	Complete report from RGI containing intermediate file data, AMR metrics, sequence information, and final AMR gene and species detection results
Intermediate files	ZIP	Files generated during AMR analysis including final AMR reports (report table and comprehensive AMR metrics), intermediate files (e.g., mapping stats and sorted BAM files), and raw outputs (e.g., report files for contigs, read mapping, and species identification data). Note: This download will also include non-host read and contig FASTA files for pipeline versions prior to AMR pipeline v1.1

Report Table vs Comprehensive AMR Metrics

Comprehensive AMR Metrics	AMR Sample Report
ARO_overall	Gene Name
Gene_family overall	Gene Family
Drug_Class_overall	Drug Class
Resistance Mechanism_overall	Resistance Mechanism
Model_overall	Model
All Mapped Reads_kma_amr	Reads
Percent Coverage_kma_amr	Read Coverage Breadth (%Cov)
Depth_kma_amr	Read Coverage Depth (Cov. Depth)
CARD*kmer prediction_kma_sp	Read Species
Cut_Off_contig_sp	Cutoff
Percentage Length of Reference Sequence_contig_amr	Contig coverage breadth (%Cov)
Best_Identities_contig_amr	Contig Percent Identity (%Id)
CARD*kmer Prediction_contig_sp	Contig Species
Contig count	Contigs

- ☐ AMR gene information
- ☐ Read metrics and results
- ☐ Contig metrics and results

Report Table -
CZ ID summary
from RGI report

**Comprehensive
AMR Metrics -**
Report from RGI

Downloading AMR Data from Project Page

Project AMR_Test2

My Data Public Upload User

Private project 1 member Share Add Data

Search My Data... Samples 3

Taxon Filters
TAXON
ANNOTATION

Metadata Filters
LOCATION
TIMEFRAME
SAMPLE TYPE

Metagenomics 2 Metagenomics - Nanopore 0 BETA Consensus Genomes 0 Antimicrobial Resistance 3

3 antimicrobial resistance

	Sample	Created On	Sample Type	Host	Passed Filters	Total Reads
☒	RR0026_watercontrol_S13_270788_reads_... User AMR_Test2	2023-05-11 21 hours ago	Slurry	Pig	243,692 99.81%	244,160
☒	RR0026_010_ASb_S10_270785_reads_nh User AMR_Test2	2023-05-10 2 days ago	Slurry	Pig	1,870,052 99.91%	1,871,686
☐	AMR_test1_reads_nh User AMR_Test2	2023-05-08 4 days ago	Wastewater	ERCC only	1,902,054 85.92%	2,213,748

2 Selected

Download icon

DESCRIPTION Edit
Testing new AMR pipeline

OVERALL

Samples 3

Projects 1

Avg. reads per sample 1,057,923

Avg. reads passing filters per sample 1,056,872

DATE CREATED

2023-05-08

Select sample(s)

AMR Project Page Download Options

Report Name	File Type	Description
Antimicrobial Resistance Results	Multiple files compressed as "tar.gz"	All results for selected samples including primary AMR report (TSV), comprehensive AMR metrics (TSV), intermediate files (BAM and TXT), and raw outputs (TXT) from CARD RGI. Note: This download will also include non-host read and contig FASTA files for pipeline versions prior to AMR pipeline v1.1
Combined AMR Results	Report file (CSV) compressed as "tar.gz"	Primary AMR report for selected samples combined into one file. The report summarizes detected AMR gene information, quality of sequence matches, and pathogen-of-origin prediction results.
Sample Metadata	One or multiple report files (CSV) compressed as "tar.gz"	User-uploaded metadata for selected samples.
Contigs	One or multiple sequence files (FASTA) compressed as "zip"	Assembled contig sequences used for AMR analysis of selected samples.

AMR Heatmaps

- Currently, CZ ID does not have heatmap functionality for the AMR module.
- You can create heatmaps outside CZ ID by running a heatmap generator script that will use the Combined AMR Report from CZ ID as input data.
 - The heatmap script can run locally on your machine or on the web via Google Colab.
 - Running the script through Google Colab is easy, with no software installations or prior coding experience needed.
 - Visit the GitHub page for the [AMR Heatmap Generator Script](#) to learn more and get started.

Notes

- May be challenging to distinguish sequencing error/artifacts from true SNPs that confer resistance
- There may be differences in AMR genes and species detected with contig data versus those identified with read data
 - KMA (reads) vs Assembly/BLAST (contigs)
 - Read AMR detection based on 1 out of 4 models (PHMs alone), whereas contig AMR detection uses 4 models
- CZ ID mNGS & AMR modules may report different species. This is due to differences in databases & alignment methods
- Known contaminant:
 - **TEM-116**: This AMR gene is part of the pUC5 plasmid, which is in a number of PCR reagents. It also may exist in clinical samples but definitely known to be in lab reagents (see [discussion](#))

Examples

- Tracing microbial contaminant in transfusion patients
- Pathogen transmission in hospital setting (neonatal surveillance)
- Pathogen dynamics within pneumonia patient (temporal study)
- AMR genes in wastewater
- Notes:
 - mNGS data: coverage breadth >5% and matches based on reads required > 5 reads mapping to AMR genes
 - WGS data: coverage breadth >50% and matches based on reads required > 5 reads mapping to AMR genes
 - Matches based on PHMs and PVMs alone

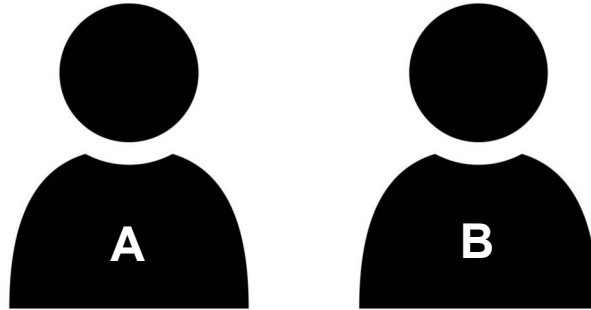
Example 1: Microbial contaminant in transfusion patients

(data from [Crawford et al 2020](#))

- Two immunocompromised patients (A and B) received platelet unit transfusions from the same donor. Both patients developed septic shock hours after the transfusion.
- Patient A treated with cefepime at the time of transfusion
- Approach: mNGS (DNA) and methicillin-susceptible *S. aureus* (MSSA) culturing followed by WGS

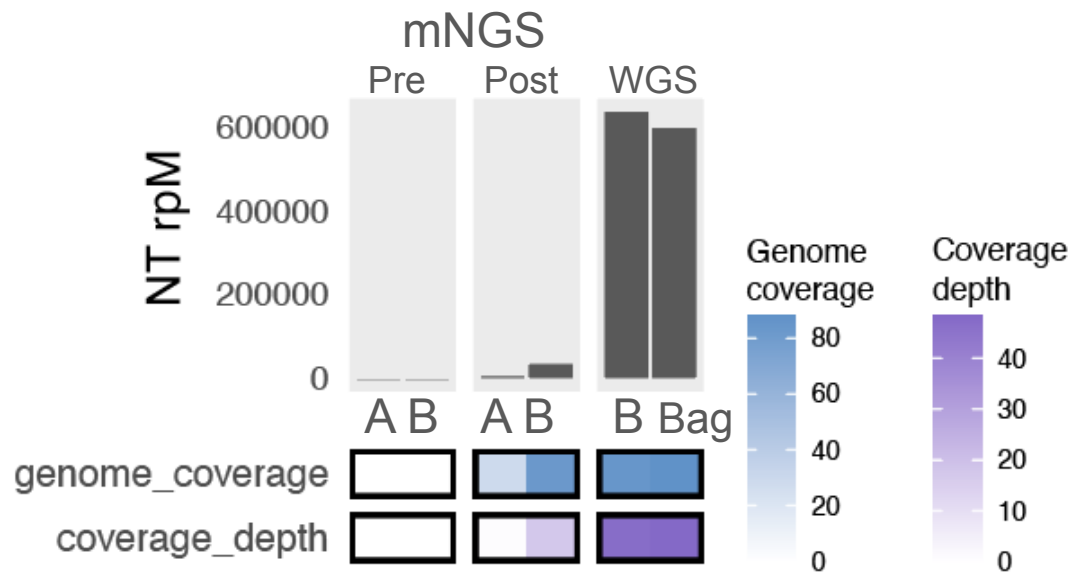


- Bag culture

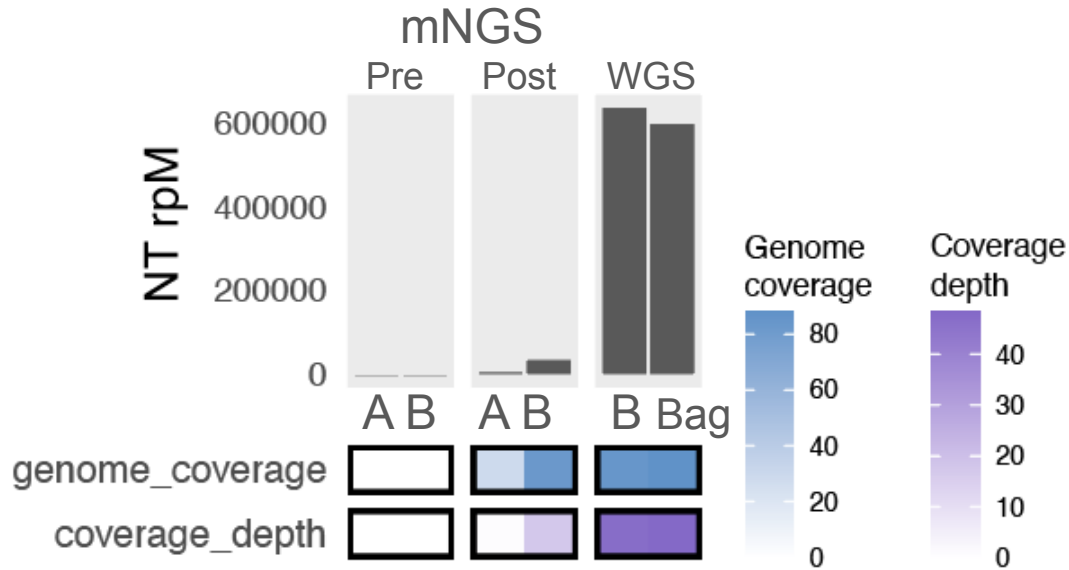


- Blood mNGS (pre- and post-transfusion)
- Blood culture (post-transfusion)

Read mapping results for *Klebsiella pneumoniae*

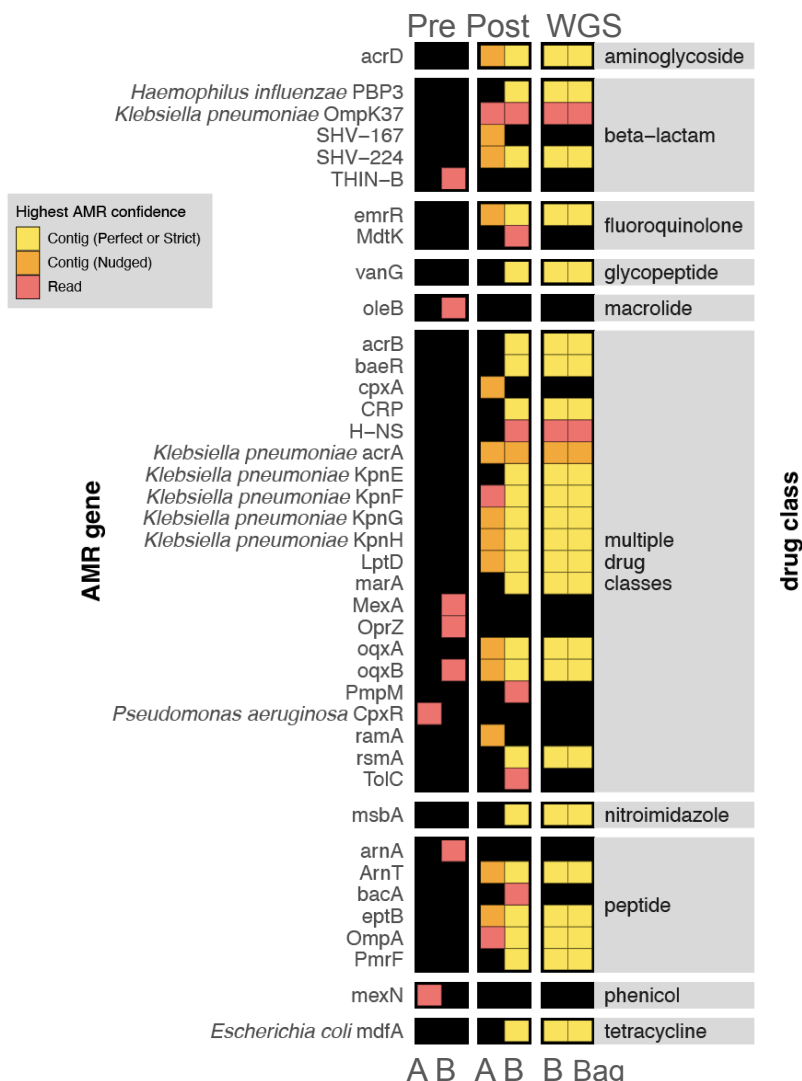


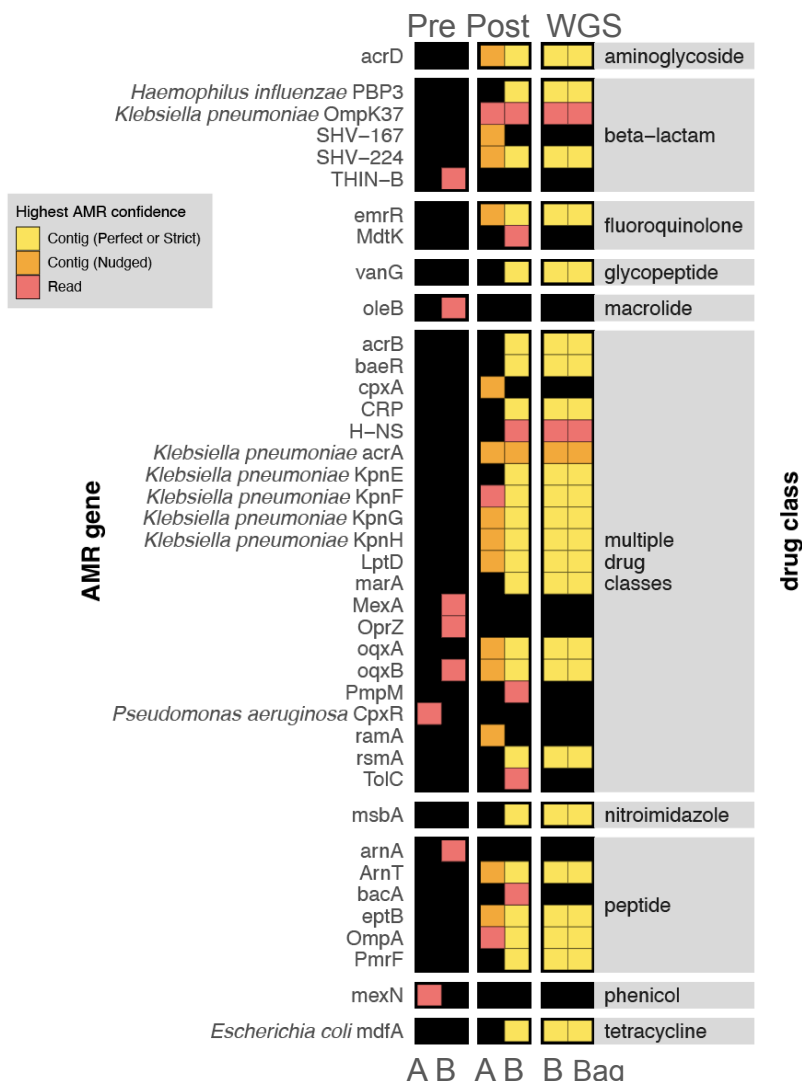
Read mapping results for *Klebsiella pneumoniae*



- Reads mapping to *K. pneumoniae* increased after treatment for both patients
- Patient A had lower levels of infection (lower rPM, no WGS data due to negative blood culture)

AMR Results





AMR Results

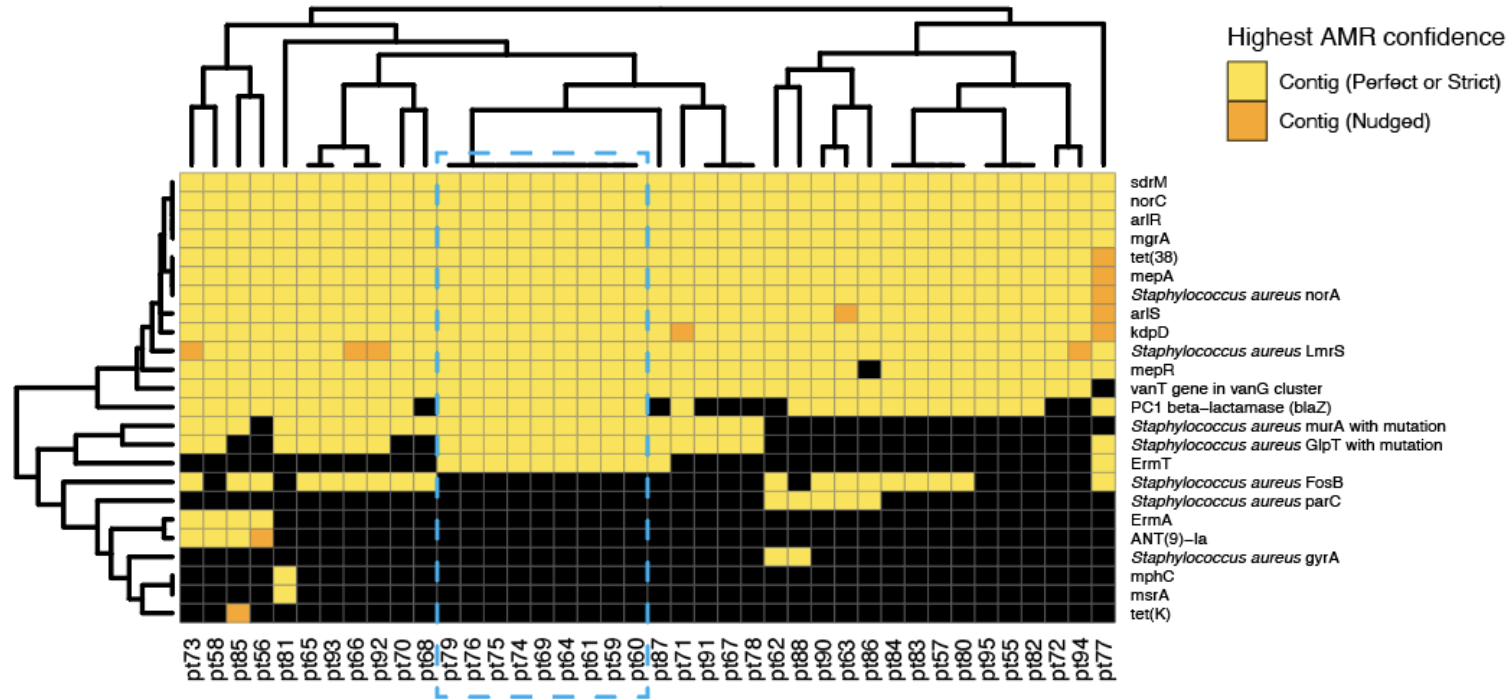
- Patient B: AMR profiles nearly identical between mNGS and WGS (culture) samples from blood and the platelet bag
- Patient A: Some *K. pneumoniae* AMR genes identified

Example 2: Pathogen transmission in neonatal unit

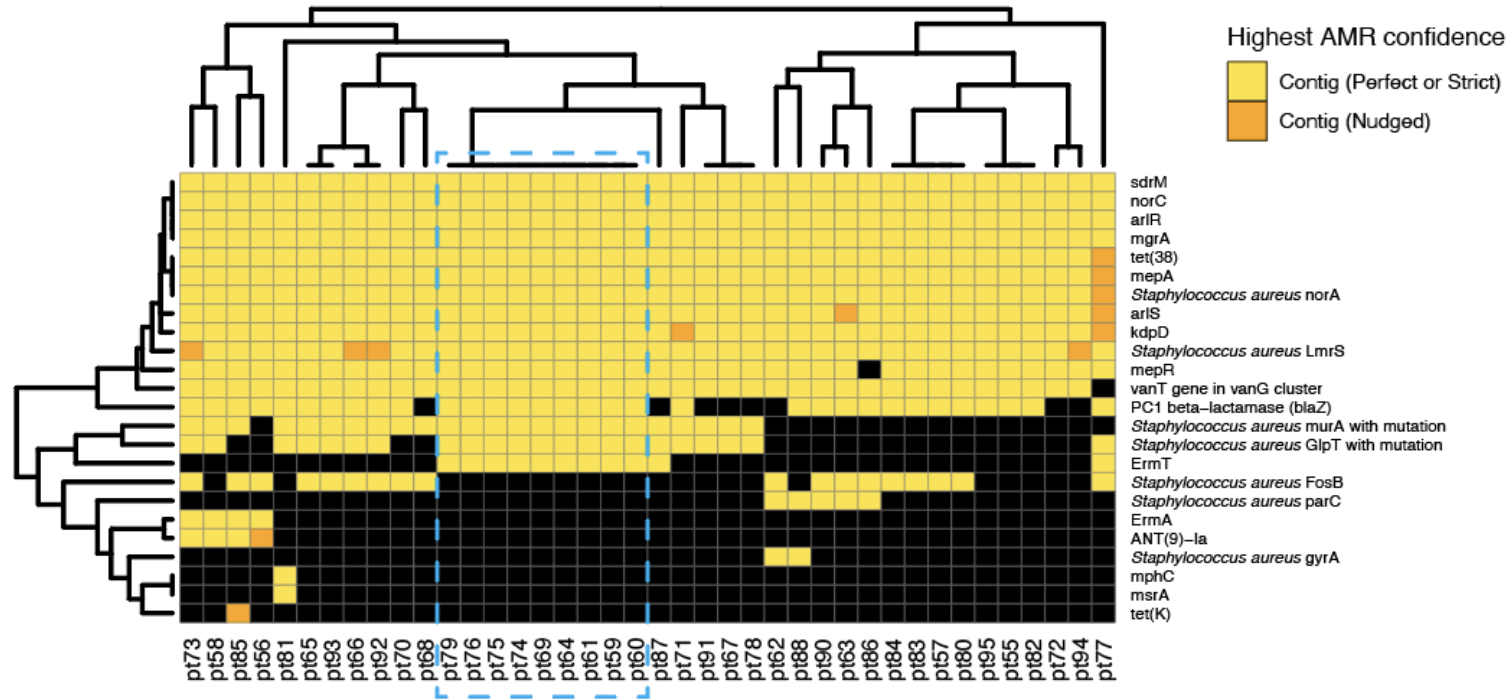
(Unpublished)

- Skin swabs collected to investigate suspected transmission of methicillin-susceptible *Staphylococcus aureus* (MSSA)
- Approach: Swabs (n = 40) used for mNGS and MSSA culturing followed by WGS

AMR profiles for MSSA isolates from patients (WGS)



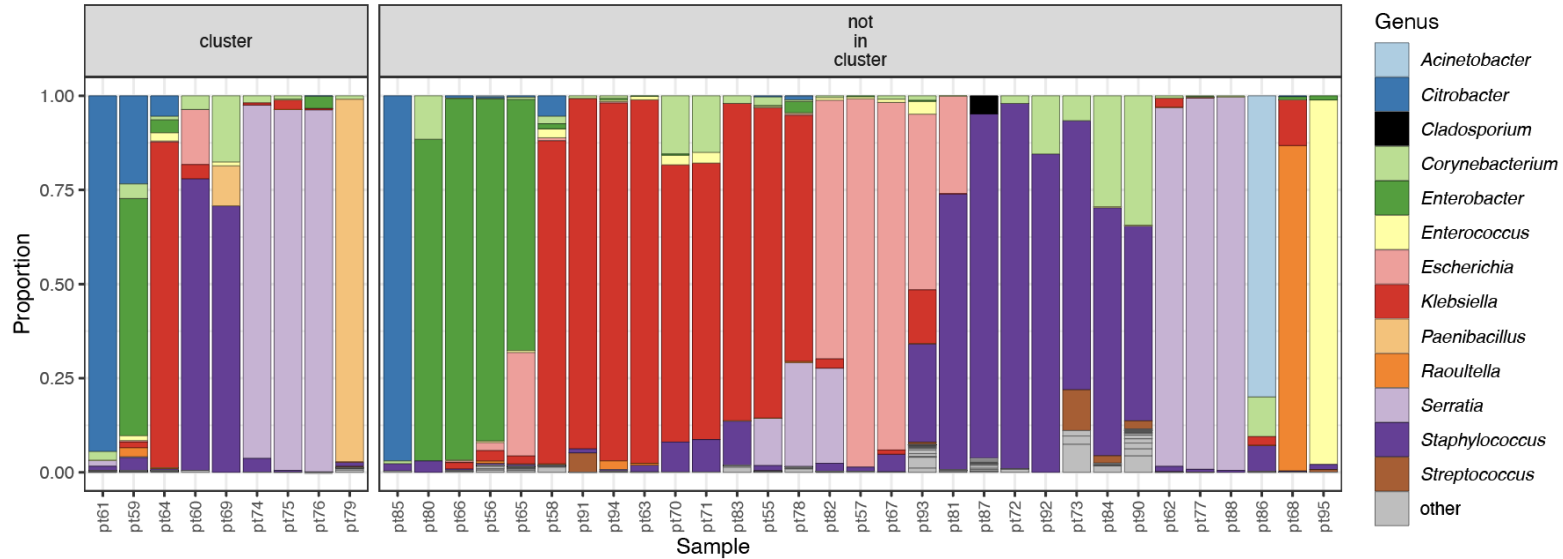
AMR profiles for MSSA isolates from patients (WGS)



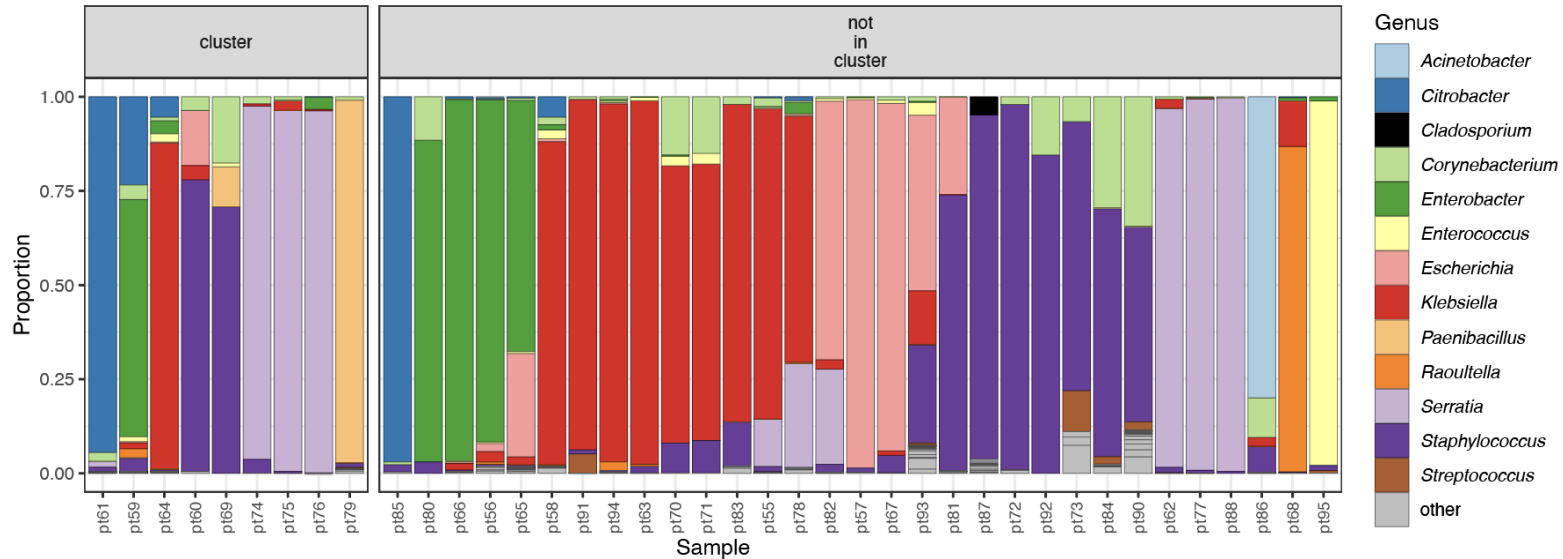
Identical AMR profiles

Downstream genome phylogenetic analysis revealed 11 SNPs between the 9 isolates

Bacterial species identified through mNGS

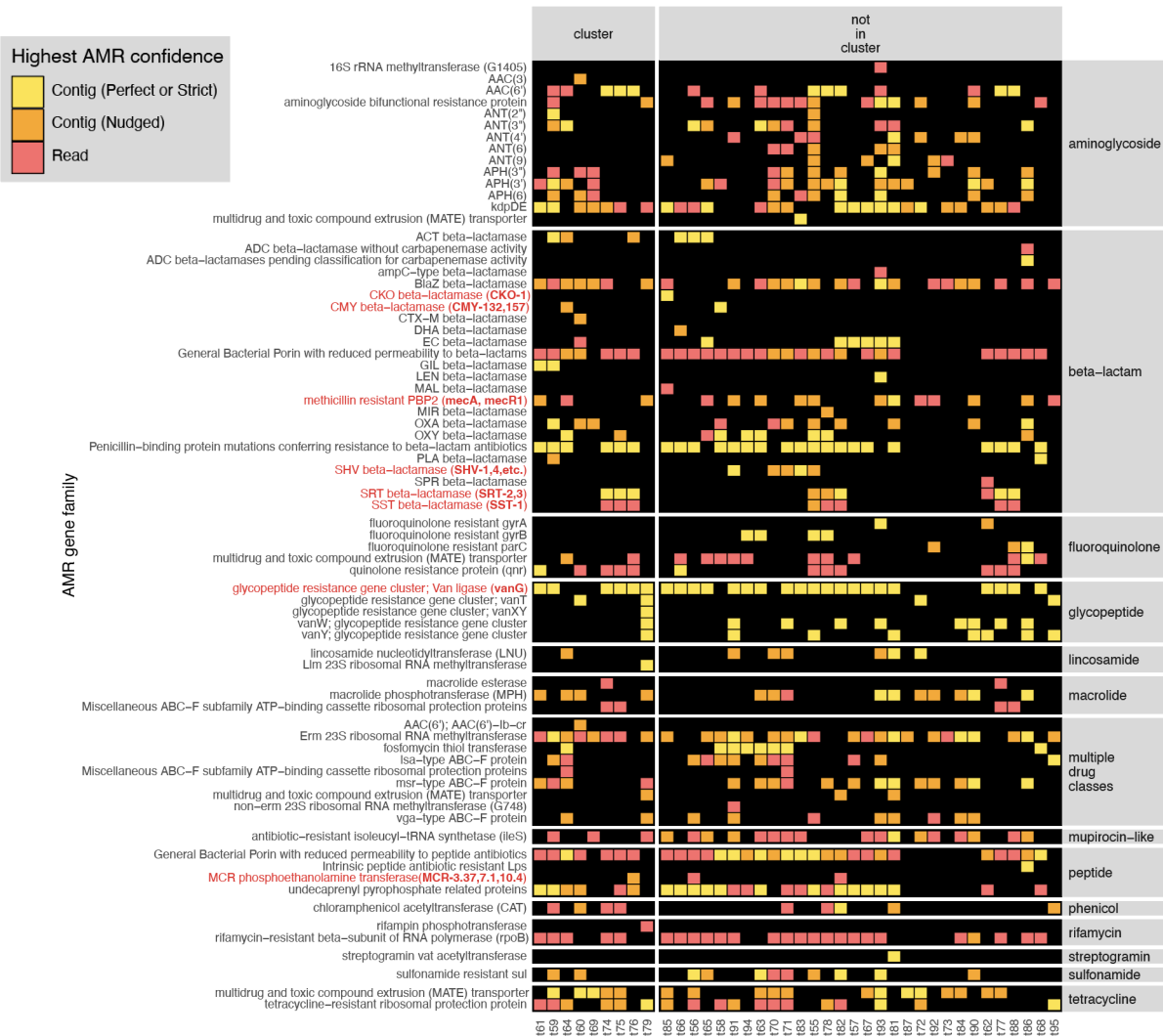


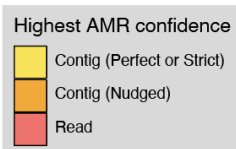
Bacterial species identified through mNGS



- Other species identified, including pathogens
- Unique profiles for samples from patients within MSSA cluster

AMR profiles for mNGS samples





AMR gene family



AMR profiles for mNGS samples

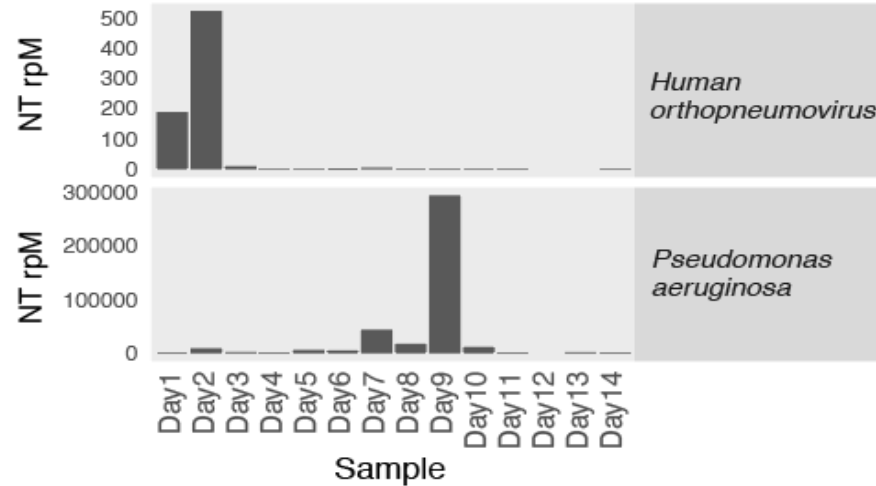
- Diverse AMR gene detection in NICU patients

Example 3: Pathogen dynamics within pneumonia patient

(Unpublished)

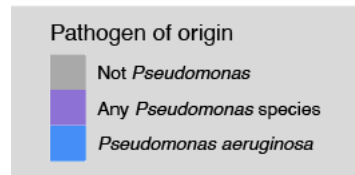
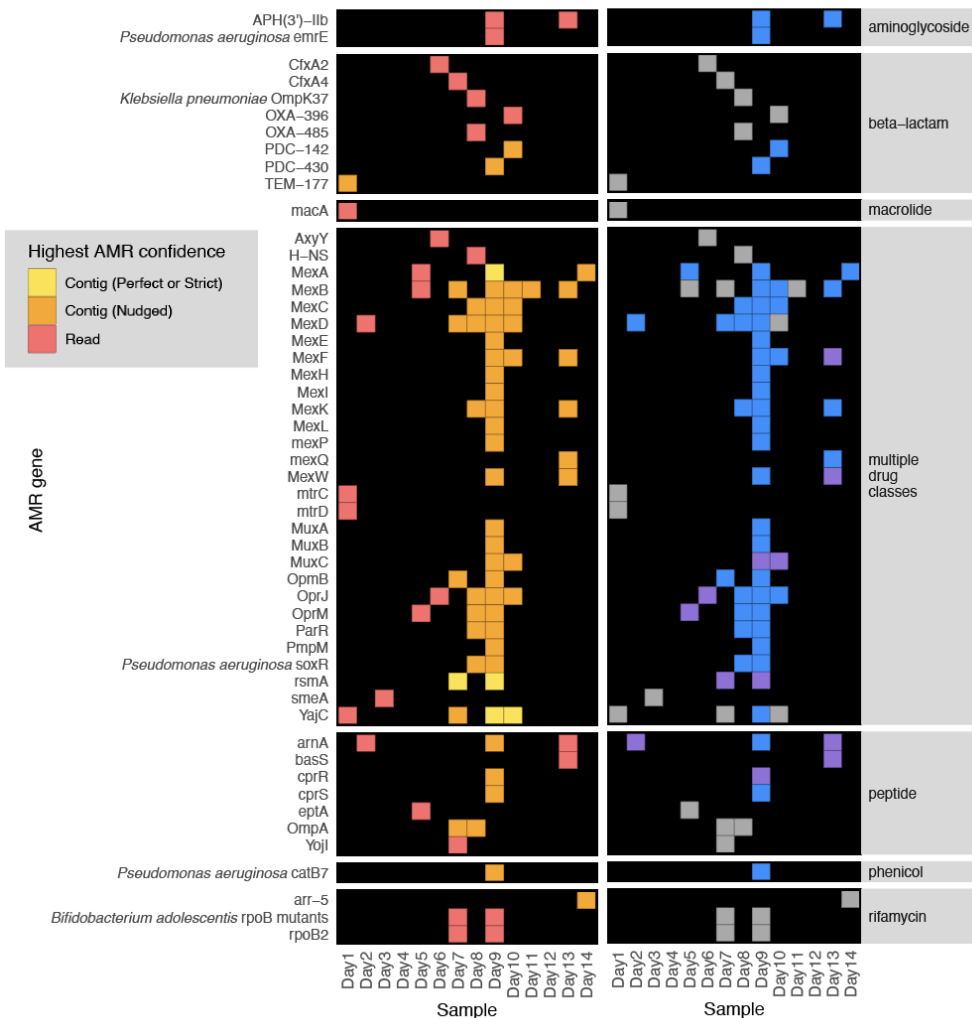
- Case: Patient intubated for severe Human orthopneumovirus (HRSV) pneumonia who subsequently developed ventilator-associated *Pseudomonas aeruginosa* infection.
- Approach: mNGS (RNA) of tracheal aspirate samples collected over a two-week period.

Read mapping results for Human orthopneumovirus and *Pseudomonas aeruginosa*

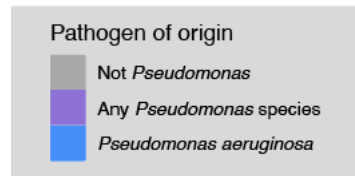
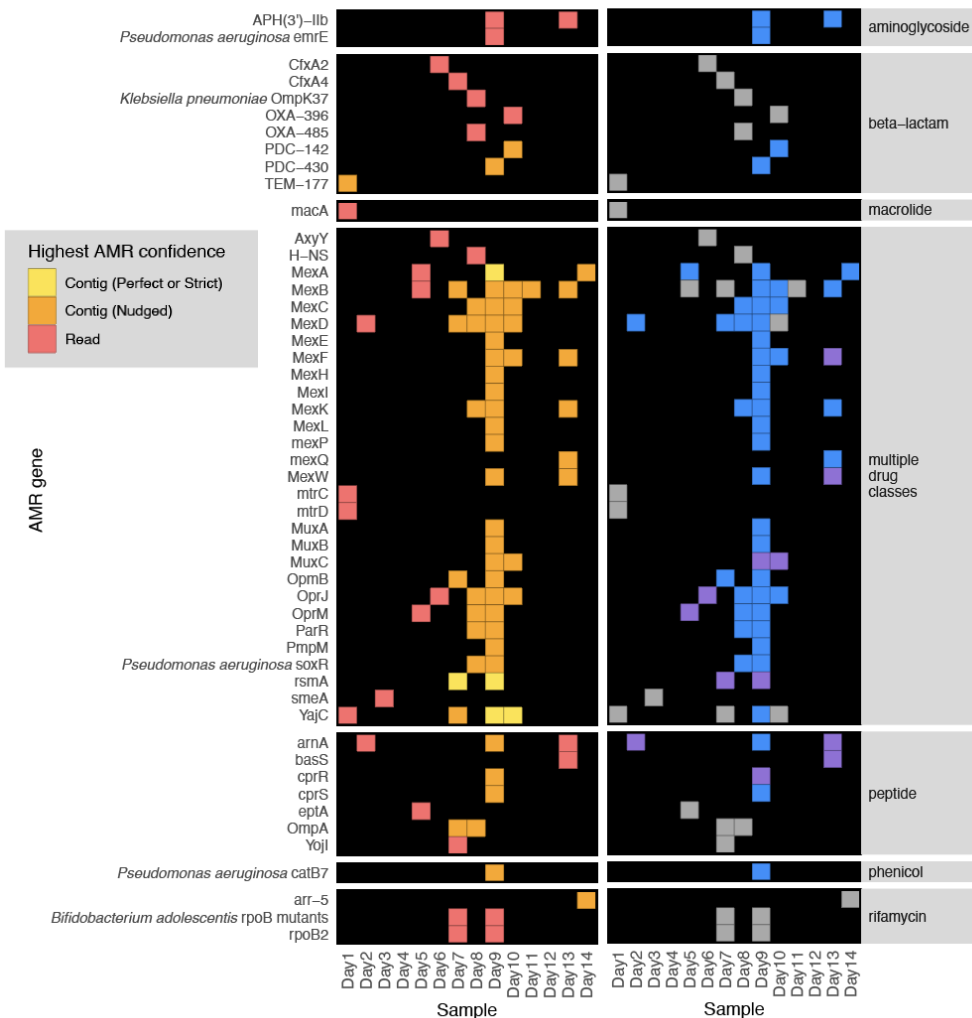


- mNGS data reflected initial viral abundance and eventual clearance followed by an increase in *P. aeruginosa* abundance

AMR Results



AMR Results

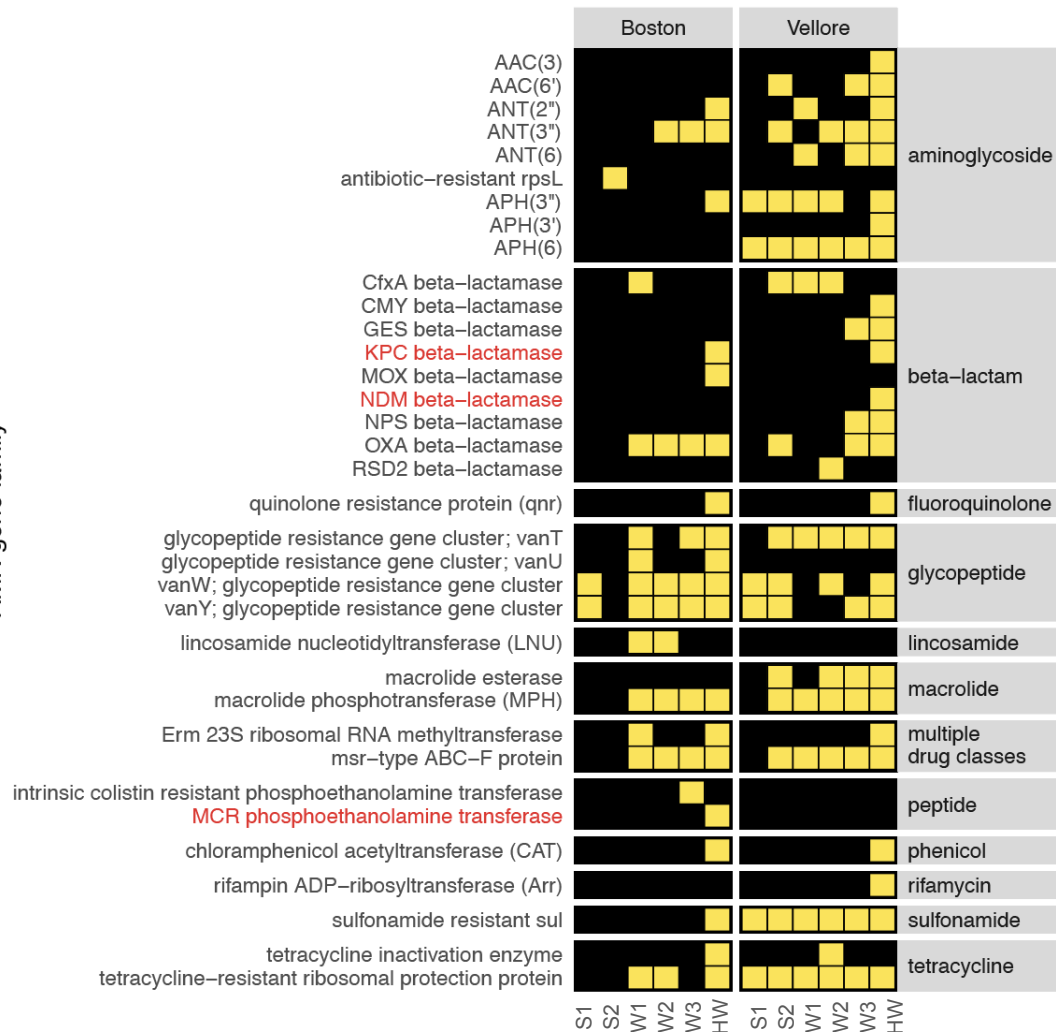


- P. aeruginosa*-associated AMR genes tracked the relative abundance of this pathogen

Example 4: AMR gene detection in wastewater

(data from [Fuhrmeister et al 2021](#))

- Wastewater surveillance study comparing Boston, USA to Vellore, India.
- Approach: mNGS (DNA) of surface water, municipal wastewater, and hospital wastewater.



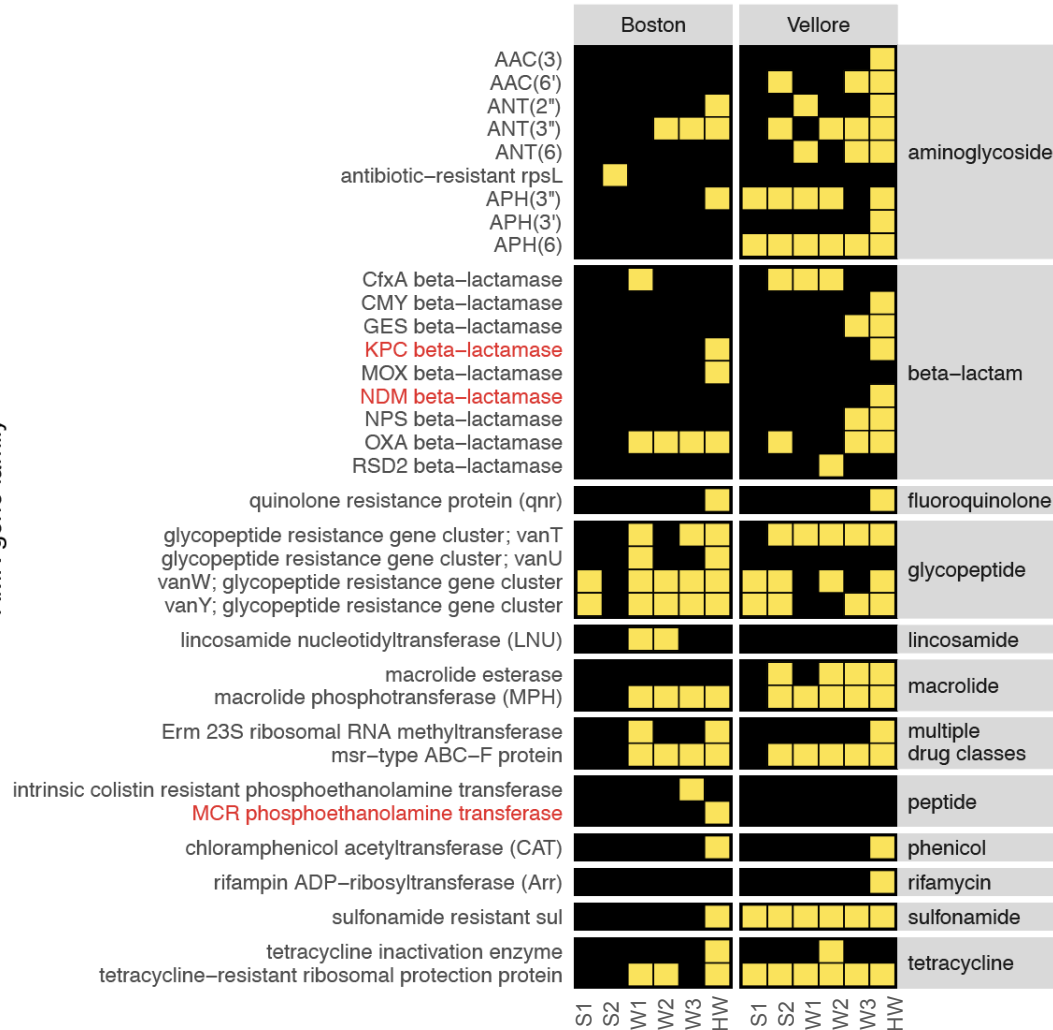
Highest AMR confidence

Contig
(Perfect or Strict)

S - Surface water

W - Municipal wastewater

HW - Hospital wastewater



Highest AMR confidence

Contig
(Perfect or Strict)

S - Surface water

W - Municipal wastewater

HW - Hospital wastewater

- Diverse AMR genes detected in wastewater.
- Hospital wastewater contains AMR genes of public health concern (infections with drug-resistant pathogens may be more prevalent in hospitals compared to the community)

Reference Guides

- [CZ ID AMR Resources](#)
 - Guides for how to upload and download AMR gene data, how to interpret and customize the AMR Sample Report, and general information about the AMR pipeline
- [Video: AMR Detection and Analysis Using CARD & RGI](#)
 - Presentation by Andrew McArthur (CARD/RGI Principal Investigator) providing background information and details about CARD and RGI workflow
- Preprint describing AMR pipeline
 - [Simultaneous detection of pathogens and antimicrobial resistance genes with the open source, cloud-based, CZ ID pipeline](#)